Code No: 136ER **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, March - 2024
ARTIFICIAL NEURAL NETWORKS
(Common to EEE, CSE)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	Compare neural network and human brain.	[2]
b)	What is error correction learning? Explain.	[3]
c)	What is feature detection?	[2]
d)	Explain the significance of learning curves.	[3]
e)	What is accelerated convergence?	[2]
f)	Explain back-propagation and differentiation.	[3]
g)	What is SOM? Explain.	[2]
h)	Write about learning vector quantization.	[3]
i)	What are called dynamic systems?	[2]
j)	Explain about attractors and their significance.	[3]

PART – B

(50 Marks)

2.a) Give a brief description on memory based learning.
b) Explain the credit assignment problem. [5+5]

OR

3.a) What is Hebbian learning? Explain.
b) Give a brief description on how neural networks are viewed as directed graphs. [5+5]

4.a) State and prove perceptron convergence theorem.
b) What is adaptive filtering problem? Explain. [5+5]

OR

5.a) Explain least mean square algorithm.
b) Describe back propagation algorithm and derive the expressions for weight update relations. [5+5]

6.a) With examples, explain about generalization and cross validation.
b) Discuss about Hessian matrix and its significance. [5+5]

OR

7.a) What are the virtues of back propagation learning? Explain.
b) Explain about supervised learning by giving examples. [5+5]

8. Explain the Kohonen's method of unsupervised learning. Discuss with example. [10]

OR

9. Discuss in detail about Adaptive pattern classification. [10]

10. Discuss in detail about neuro-dynamical models. [10]

OR

11. With a neat sketch, describe the architecture of Hopfield network models. [10]

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